AudioExplorers

The DSP Case
Galactic OAE Retrieval Mission

Background & Introduction

General Introduction to TEOAEs

Transient Evoked Otoacoustic Emissions (TEOAEs) are weak acoustic signals generated by the outer hair cells of the cochlea in response to short acoustic stimuli such as clicks. They are widely used in audiology because:

- Healthy outer hair cells actively amplify sound inside the cochlea, and this process produces measurable echoes.
- These echoes travel backwards through the middle ear and can be recorded in the ear canal.
- If no TEOAE is present, it typically indicates outer hair cell dysfunction and is often associated with hearing loss.

Demant

The classical method involves presenting a rapid sequence of acoustic transients and averaging the resulting ear canal response.

The **nonlinear +1+1+1-3 stimulus paradigm** (three positive clicks followed by one negative triple amplitude click) is commonly used to suppress linear components from the middle ear, isolating the cochlear generated nonlinear response.

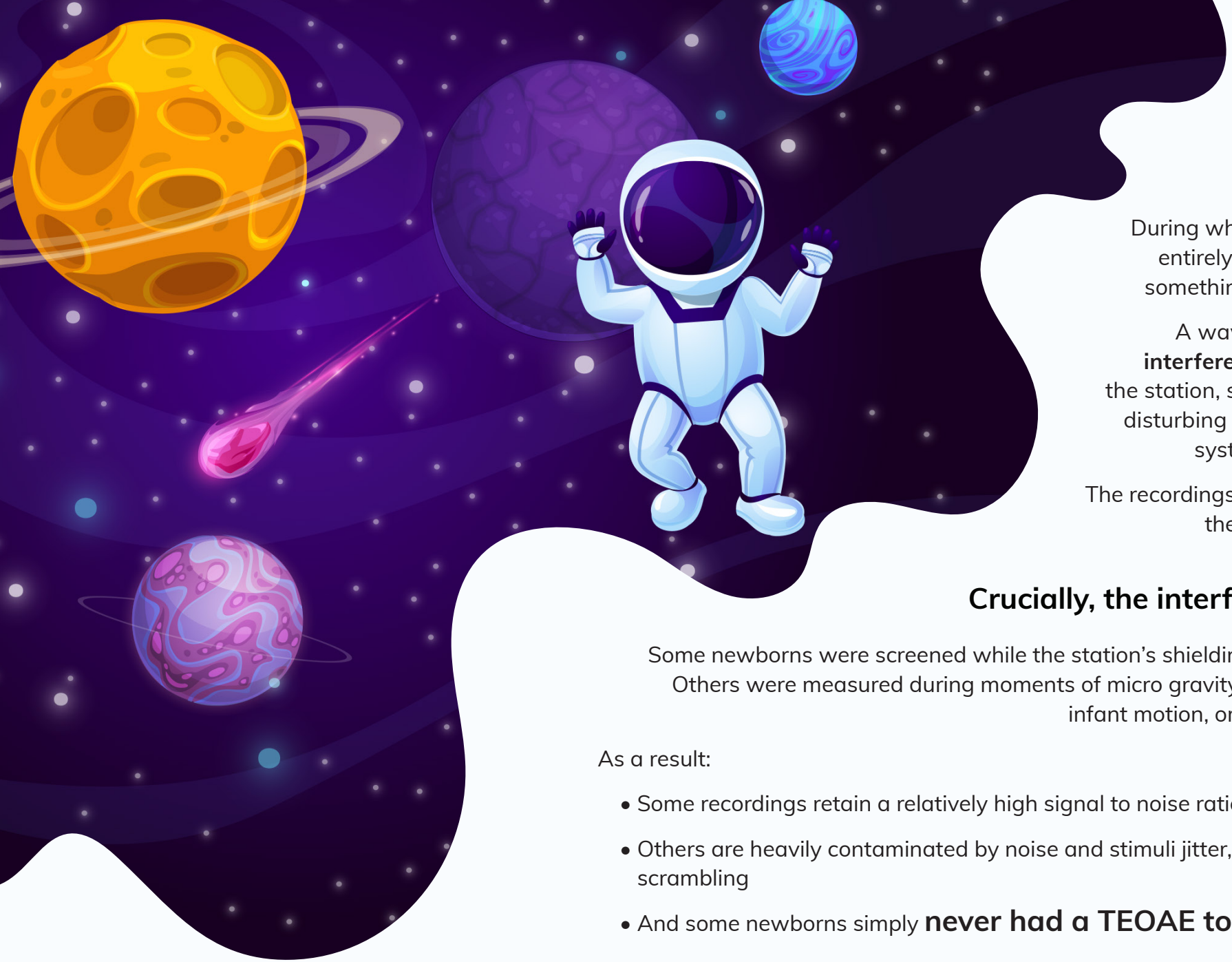
For additional background, there is an excellent introductory playlist on [YouTube:]

The challenge

The year is **2126**.

At the heart of the interstellar research station **DGS Titanius-12** lies one of the galaxy's most advanced **newborn hearing screening units**.

Every newborn arriving on the station is routinely screened using **Transient Evoked Otoacoustic Emissions (TEOAEs)** - a fast, objective, and non invasive way to verify healthy cochlear function at the very start of life.



During what should have been an entirely routine screening cycle, something unexpected happens:

A wave of **stochastic cosmic interfere-rence** sweeps through the station, subtly but unpredictably disturbing the auditory acquisition systems mid measurement.

The recordings are not destroyed - but they are no longer pristine.

Crucially, the interference is uneven!

Some newborns were screened while the station's shielding was stable and calm. Others were measured during moments of micro gravity drift, probe movement, infant motion, or surges of cosmic noise.

As a result:

- Some recordings retain a relatively high signal to noise ratio
- Others are heavily contaminated by noise and stimuli jitter, polarity and order scrambling
- And some newborns simply **never had a TEOAE to begin with**

After the incident, engineers discover the following:

- **12 newborn (patient) recordings**, each affected differently by stochastic interference and.
- Only **8 of these recordings contain a genuine TEOAE response**
- A secure backup file, **lostOaes.json,** holding the 8 true OAE responses - but now shuffled and unlabeled
- A nonlinear **+1+1+1-3** TE-OAE stimulus paradigm was used throughout, encoded via trigger types **A, B, C, and D**

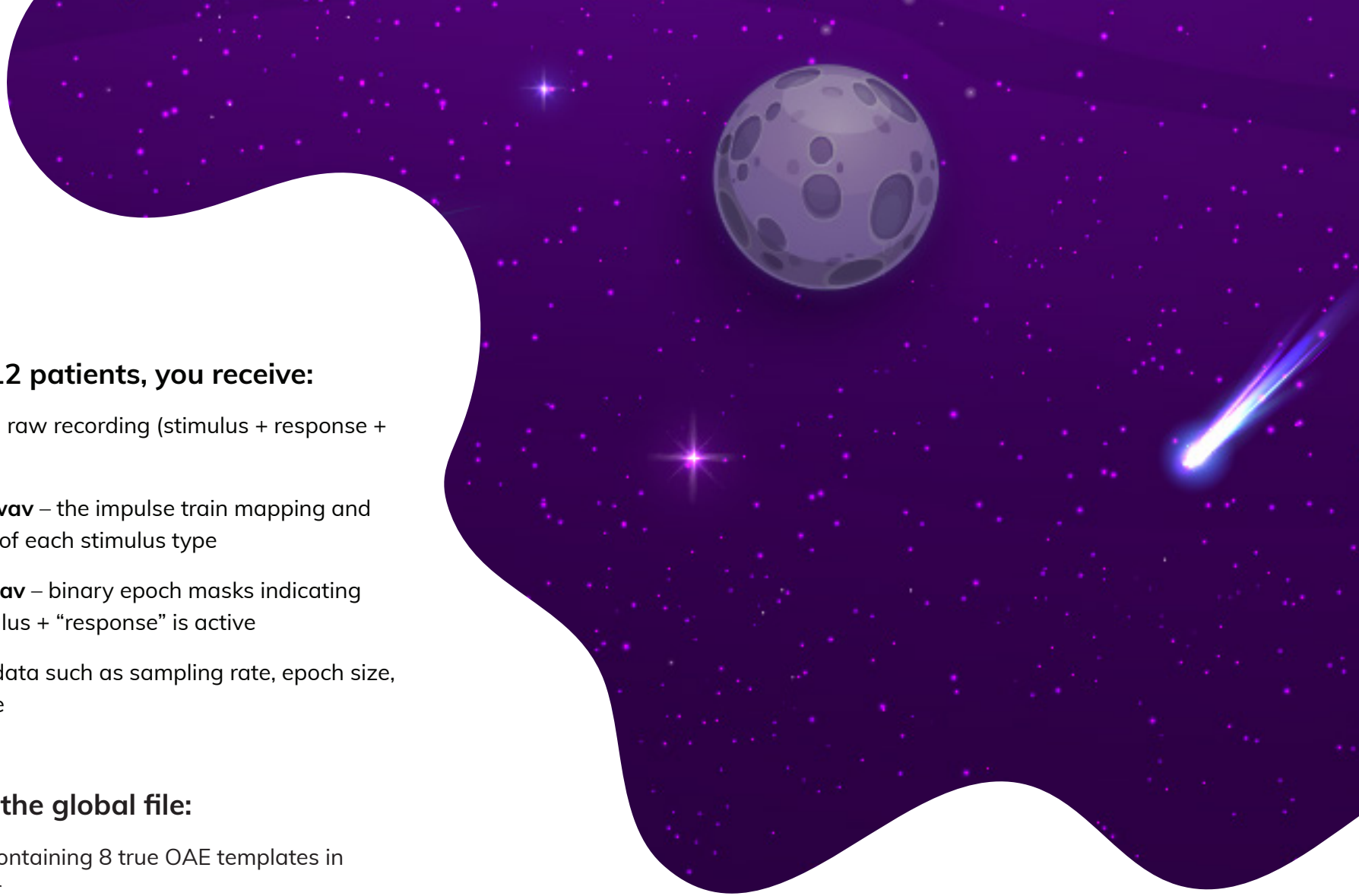
The screening unit can no longer automatically decide OAE present or not.

That decision is now up to **you!**

Important note from mission Control:

Due to experimental signal generators, creative engineering decisions, and questionable cosmic shielding, **both the recorded signals and the scrambling mechanisms applied to them are entirely artificial**. They are designed to challenge signal processing algorithms rather than to obey real world physics.

The waveforms, trigger behavior, polarity tricks, and scrambling logic should not be interpreted as representative of real clinical TEOAE recordings - on Earth or anywhere else in the galaxy.



The case

For each of the 12 patients, you receive:

1. **rec.wav** – the full raw recording (stimulus + response + noise)
2. **triggerA/B/C/D.wav** – the impulse train mapping and polarity for start of each stimulus type
3. **activeA/B/C/D.wav** – binary epoch masks indicating when each stimulus + “response” is active
4. **info.json** – metadata such as sampling rate, epoch size, and stimulus rate

You also receive the global file:

- **lostOaes.json** – containing 8 true OAE templates in randomized order

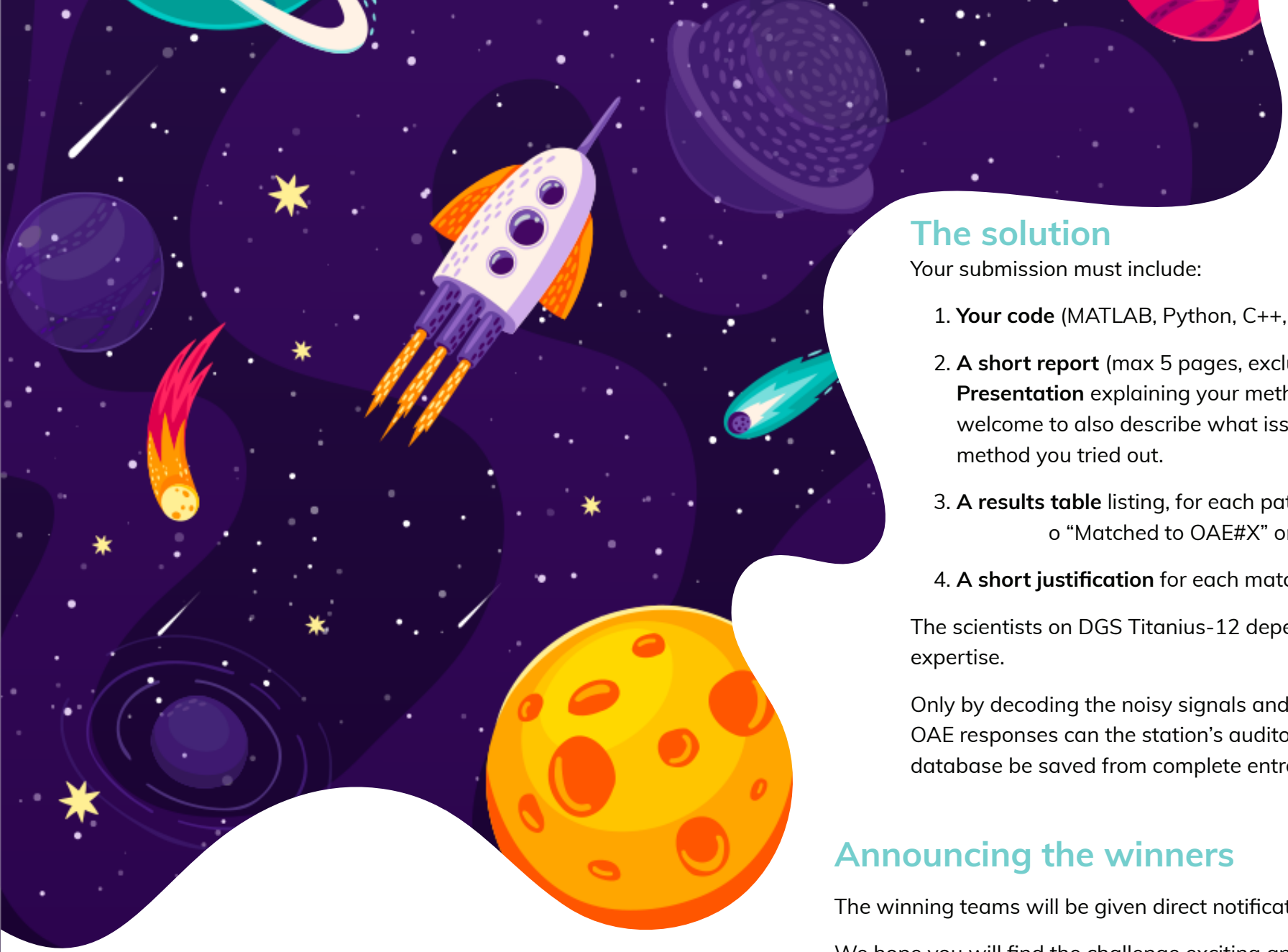
Only 8 out of 12 patients have an actual OAE.

Your algorithm must also mark which of the 4 patients with no OAE present.

Your task is to recover the cochlear responses, identify which newborns truly exhibit OAEs, and correctly match the recovered responses to the lost templates - all while navigating the uneven effects of cosmic interference.

Your mission is to:

- **Reconstruct each patient's TEOAE response** using the four signals for A, B, C, D.
- **Determine whether the patient contains a genuine OAE response (pass)** or is OAE negative (refer).
- **Match each of the 8 clean OAE templates in lostOaes.json** to the correct patient recording - and justify your choices.
- **Clearly explain why your method works** and how you deal with the cosmic stochastic interference.



The solution

Your submission must include:

1. **Your code** (MATLAB, Python, C++, etc.)
2. **A short report** (max 5 pages, excluding front page) or **Video Presentation** explaining your method. You are more than welcome to also describe what issues you encountered and method you tried out.
3. **A results table** listing, for each patient:
   - o “Matched to OAE#X” or “No OAE present”
4. **A short justification** for each match

The scientists on DGS Titanius-12 depends on your engineering expertise.

Only by decoding the noisy signals and recovering the eight true OAE responses can the station’s auditory hearing screening database be saved from complete entropy.

Announcing the winners

The winning teams will be given direct notification via email or phone.

We hope you will find the challenge exciting and fun to work on, and we look forward to seeing your solutions.

Greetings, and may the signal to noise ratio be ever in your favor.

The Audio Explorers Team